

Kesher Trust-Forward Personality Stack on Google AI Studio and Gemini

Verdict and Scientific Contract

Executive Verdict

A trust-forward Big Five Aspects (BFAS) system can be productized for dating *without* becoming creepy or deterministic if you treat BFAS as a **self-reflection and communication scaffolding layer**, not a hidden ranking oracle. The implementation contract is:

- Trait measurement is **deterministic, auditable, user-owned, and opt-in** (LLMs do **not** score BFAS). **[VERIFIED]** BFAS is a 100-item measure yielding 10 aspects nested within 5 Big Five domains. ¹
- Trait interpretation is **bounded**, framed as **tendencies**, and carries uncertainty markers (no fate, no diagnosis, no “AI knows you”). **[INFERRED]** This aligns with relationship-personality evidence showing meaningful but non-deterministic associations (e.g., Neuroticism relates negatively to partner satisfaction in meta-analytic work). ²
- Compatibility is **reflective, not predictive**: a “compatibility reflection” should explain *possible friction points and strengths* and suggest *questions and micro-habits*, rather than outputting a synthetic “score.” **[INFERRED]** Actor and partner trait effects matter; similarity effects tend to be smaller/less consistent than people assume. ³
- Google-native advantage is **structured outputs + grounding (Maps/Search) + thinking controls + Firebase AI Logic security controls**, mapped into calm, premium UX, with citations where grounding is used. **[VERIFIED]** Google documents structured output via JSON Schema, grounding metadata for citations, Maps grounding attribution requirements, and thinking controls. ⁴

From the provided Kesher repo, the product tone and constraints are already aligned with “serious, calm, non-objectifying” rules (e.g., “no attractiveness scoring,” “no photo inference,” “no autosend,” “use only whitelisted signals”), and the app already uses **server-side Gemini calls only** with structured schemas and a feature allowlist. **[VERIFIED]** ⁵ ⁶ ⁷

Canonical Scientific Model Summary

Canonical model (as required):

- Big Five domains: Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness to Experience.
- Ten aspects:
 - Extraversion: Enthusiasm, Assertiveness
 - Neuroticism: Withdrawal, Volatility
 - Agreeableness: Compassion, Politeness
 - Conscientiousness: Industriousness, Orderliness

- Openness to Experience: Openness, Intellect

This “10 aspects” layer is supported by factor-analytic work finding two separable (correlated) subfactors within each Big Five domain and describing the creation/validation of BFAS. [VERIFIED] 1

BFAS is explicitly the basis of Understand Myself 8, which also offers a “relationship/couple report” concept (interpretation-only here; not a validity proof). [VERIFIED] 9

Cross-cultural evidence supports that the Five-Factor Model structure replicates across cultures even when mean levels differ and interpretation requires caution. [VERIFIED] 10

Ethics baseline: the Standards for Educational and Psychological Testing (joint product of American Educational Research Association 11, American Psychological Association 12, and National Council on Measurement in Education 13) are open access and positioned as a major guidance reference for sound and ethical test use. [VERIFIED] 14

Product Thesis Translation

Thesis: “Personality makes dating calmer when it is *owned by the user* and used to improve (a) self-presentation, (b) communication, and (c) expectation management—not to ‘grade’ people.”

Direct translation into a contract:

- **Measurement contract (deterministic):**
 - BFAS intake → aspect scores → domain scores computed as defined by the instrument. [VERIFIED] BFAS yields the 10 aspects and Big Five domain-level scores (as means of aspect pairs in the original work). 1
 - LLM role: language explanation only (no scoring; no changing scores). [HEURISTIC]
- **Interpretation contract (LLM-bounded):**
 - Output: “tendencies,” “in some situations,” “may,” and “watch-outs.” [INFERRED] This matches relationship evidence: personality relates to relationship satisfaction but does not deterministically predict outcomes. 2
- **Compatibility contract (reflection, not prediction):**
 - Output: “conversation prompts,” “possible friction,” “micro-habits,” never “compatibility score.” [INFERRED] Similarity effects are not the main story compared to actor/partner trait effects. 3
- **Trust contract (provenance + disclosure):**
 - Any user-facing personalization must cite which **user-visible inputs** were used (via a provenance ledger). [INFERRED] This is the product translation of Kesher’s existing “whitelisted signals only” stance in match explanations. 5 15

Best Google AI Studio / Gemini Wedge

Smallest uniquely “Google-native” wedge that still feels like real substance:

Wedge = Personality Reflection (BFAS) + “Why this match” provenance + Grounded Date Concierge (Maps grounding with citations) + Rewrite-first Messaging Coach

Why it wins:

- Personality reflection is “high value, low creep”: user learns about themselves, not “ranked by AI.” **[INFERRED]**
- “Why this match” provenance is a trust differentiator (Kesher already enforces whitelisted signal discipline in system instructions). **[VERIFIED]** ⁵
- Maps grounding provides safe, factual, local date planning with required source attribution and optional widgets—harder to replicate cleanly on other stacks. **[VERIFIED]** ¹⁶
- Rewrite-first messaging support aligns with “human remains the author” (and the repo already emphasizes “drafts only,” no identity invention). **[VERIFIED]** ⁵

Gemini and Google Platform Decisions

Capability-by-Capability Verdict

The table below assigns roles for each required AI Studio / Gemini / Firebase surface, with “now/later/internal/never” determinations.

Legend:

- Member-facing now = safe + high leverage for MVP
- Member-facing later = valuable but needs stronger governance or norms data
- Internal ops only = useful, but should not touch member UX directly
- Prototype-only = accelerates demos, but too risky / unstable for production
- Do not use = violates non-negotiables or high creepiness risk

Capability (required list)	Verdict	Why (contract-first)	Evidence label
Add Gemini intelligence	Member-facing now	Core requirement; must be bounded by schemas + provenance.	[HEURISTIC]
Structured outputs	Member-facing now	UI safety + determinism: schemas reduce hallucinated formats; supported via JSON Schema subset. ¹⁷	[VERIFIED]
Safety settings	Member-facing now	Baseline harm filtering per category and threshold controls. ¹⁸	[VERIFIED]
Function calling	Member-facing later (limited) + Internal ops now	Best for DB/tool bridging; but increases attack surface. Should be server-only with strict allowlists. ¹⁹	[VERIFIED]
Google Maps grounding	Member-facing now (date planning only)	Differentiator; requires attribution + source UI; tool is explicit and has usage requirements. ¹⁶	[VERIFIED]

Capability (required list)	Verdict	Why (contract-first)	Evidence label
Google Search grounding	Member-facing now (safety center, policies, travel ideas)	Must surface citations; useful for safety FAQ and time-sensitive info. ²⁰	[VERIFIED]
Analyze images	Internal ops only (moderation / profile QA)	Member-facing image “personality” inference is disallowed; optional low-risk uses: detect prohibited content, check clarity—still sensitive. ²¹	[INFERRED]
Enable high thinking	Member-facing now (only for reflection/ compatibility/premium)	Use thinking controls for depth vs latency; Gemini docs define thinkingLevel/budget. ²²	[VERIFIED]
Add low-latency responses	Member-facing now	Use Flash/Flash-Lite + low thinking for chat assist. Models page positions Flash(-Lite) for low latency. ²³	[VERIFIED]
AI Studio Build mode	Prototype-only	Great for rapid iteration; not a production governance boundary. Build mode supports deploy/export. ²⁴	[VERIFIED]
Annotation mode	Prototype-only	Excellent for UI iteration; not a runtime feature. ²⁴	[VERIFIED]
App Gallery / remix	Prototype-only	Inspiration + cloning; not prod. ²⁴	[VERIFIED]
AI Chips	Prototype-only	Prompt-time feature toggles for build; not prod governance. ²⁴	[VERIFIED]
Code editor / export paths	Prototype → Production bridge	Export to GitHub is explicitly supported from Build mode. ²⁴	[VERIFIED]
Cloud Run deploy path	Member-facing later (staging)	Good for early production; requires your own auth, logging controls, and policy enforcement. AI Studio mentions Cloud Run deploy. ²⁴	[VERIFIED]
Firebase AI Logic	Member-facing now (select features)	Client SDK access to Gemini/Imagen with App Check integration. ²⁵	[VERIFIED]
App Check	Member-facing now	Protects model APIs from unauthorized clients; AI Logic explicitly recommends App Check early. ²⁶	[VERIFIED]
per-user rate limits	Member-facing now	AI Logic provides per-user quota mechanism; crucial for abuse control. ²⁷	[VERIFIED]

Capability (required list)	Verdict	Why (contract-first)	Evidence label
Remote Config / model routing	Member-facing later	Useful for feature flags + routing + safe rollouts; Remote Config warns not to store confidential data. ²⁸	[VERIFIED]
Gemini API direct backend usage	Member-facing now (server-side for sensitive)	Best for high-control server enforcement; repo already uses server-side only pattern. ⁶ ⁷	[VERIFIED]
Vertex AI	Member-facing later (only if justified)	Use only if you need enterprise controls/data residency; otherwise adds complexity. Firebase AI Logic supports Vertex provider. ²⁹	[INFERRED]
Code execution	Internal ops only	Useful for evaluation simulation; not member-facing. Code execution tool exists and has limits/IO details. ³⁰	[VERIFIED]
Logs and datasets	Internal ops only	Powerful but risky: logs retention + sharing are explicit; must be gated/opt-in and minimized. ³¹	[VERIFIED]
Files API	Member-facing later (voice journaling) + Internal ops now	Enables reuse of files; but increases retention risk for sensitive dating data. ³²	[INFERRED]
File Search	Internal ops only	Use for internal case review bundles; avoid member-facing “upload your chat for analysis” flows.	[HEURISTIC]
Interactions API	Prototype-only → Internal ops later	Default storage + retention (1 day free, 55 days paid) is unsuitable for sensitive dating unless store=false everywhere. ³³	[VERIFIED]
Live API	Member-facing later (premium reflection only)	Real-time audio is compelling but high-risk; Live tool support matrix exists. ³⁴	[VERIFIED]
Antigravity	Prototype-only / internal build acceleration	Officially positioned as agentic dev platform; not a member-facing capability. ³⁵	[VERIFIED]

Product Flow Design

Product Surface Map

Below maps the capability stack onto the required surfaces (A–L). Each surface includes: user goal, capability, visibility, speed/depth, trust risks, privacy risks, and free vs premium boundary.

A. Onboarding

User goal: understand “what this is” and why BFAS helps without feeling evaluated.

Capability stack:

- Deterministic UI + minimal Gemini copy (structured output optional).
- Optional “Daily Picks intro” bilingual calm tone (repo already generates English/Hebrew intros).

[VERIFIED] 7

Visibility: Visible but low-key (“assistant voice” as brand tone).

Fast vs deep: Fast (low thinking). **[VERIFIED]** thinking controls exist; use low for onboarding. 22

Trust risks: Overclaiming “science says you’re X.”

Privacy risks: Collecting personality before consent.

Correct boundary:

- Free: explanation + opt-in toggle
- Premium: none (dignity is not paywalled) **[HEURISTIC]**

B. Personality assessment intake

User goal: take a valid assessment without fatigue; understand that it’s optional and reversible.

Capability stack:

- Deterministic item delivery + scoring (no LLM).
 - LLM: microcopy only (break reminders, encouragement) bounded.
 - Optional short form pathway (BFAS-40 exists in some validations; Hebrew validation unknown).
- [VERIFIED]** BFAS is 100 items; aspect structure is defined. 1 **[VERIFIED]** BFAS and BFAS-40 validations exist (non-Hebrew). 36 **[UNKNOWN]** validated Hebrew BFAS/BFAS-40 norms/invariance.

Visibility: Visible; explicitly labeled as self-report tendencies.

Fast vs deep: No thinking; deterministic.

Trust risks:

- “One true you” framing.
- “You can only take once” like Understand Myself (they restrict retakes; that’s their product, not automatically correct for Keshet). **[VERIFIED]** Understand Myself states single administration constraint. 37

Keshet recommendation: allow retake/reset but label “retakes can change results; treat as a snapshot.”

[INFERRED]

Privacy risks:

- Raw item responses are the most sensitive field (more than derived aspect scores). **[HEURISTIC]**

Free vs premium:

- Free: full assessment (do not paywall assessment)
- Premium: advanced reflection modules, not the measurement itself. **[HEURISTIC]**

C. Profile creation / profile editing

User goal: present self authentically, especially in Hebrew-first context.

Capability stack:

- Structured output “bio coach” (repo already does Hebrew-first drafts; server-side only; no invented facts). **[VERIFIED]** 5 38 15
- Add personality-aware suggestions **only if user opts in**, and only as “phrasing options” (e.g., how to describe low Assertiveness without self-judgment). **[INFERRED]**

Visibility: Visible; always “drafts” and “options.”

Fast vs deep: Mostly fast (low thinking).

Trust risks: Stereotyping observance/politics; repo already forbids. **[VERIFIED]** 5

Privacy risks: leaking private personality into public profile by default.

Free vs premium:

- Free: Bio coach baseline
- Premium: “values/observance phrasing assistant” with more iterations + tone packs. **[HEURISTIC]**

D. Personality reflection report

User goal: understand self patterns relevant to dating; feel seen, not judged.

Capability stack:

- Deterministic scores + LLM explanation with structured outputs (cards).
- Use high thinking for nuance, but bounded by schema and “no diagnosis.” **[VERIFIED]** thinking controls exist. 22

Visibility: Visible; user-owned; export/delete.

Fast vs deep: Deep (high thinking) but asynchronous UX (loader + “calm pacing”). **[HEURISTIC]**

Trust risks:

- “clinical” tone
- Overprecision (“your marriage will fail because...”)

Evidence base:

- Big Five relate to relationship satisfaction in meta-analysis; Neuroticism is a consistent negative correlate; Agreeableness and Conscientiousness show positive associations. **[VERIFIED]** ³⁹
- But relationship outcomes are multi-factorial; therefore the report must remain reflective. **[INFERRED]**

E. Couple / compatibility comparison report

User goal: understand why a connection may feel easy/hard; get concrete questions to discuss.

Capability stack:

- Structured output compatibility cards that only use:
 - 1) each person's opted-in trait disclosures
 - 2) each person's self-stated relationship preferences
- Emphasize actor/partner effects; downweight similarity. **[VERIFIED]** dyadic evidence considers actor/partner/similarity. ⁴⁰

Visibility: Visible only after mutual opt-in + mutual share.

Fast vs deep: Deep (high thinking) but with strict uncertainty markers.

Trust risks: "compatibility score," "soulmate framing," hidden ranking.

Privacy risks: revealing one partner's traits without consent.

Free vs premium:

- Free: 1–2 compatibility reflection cards + questions
- Premium: expanded "micro-habits plan," conflict scripts, and post-date reflection journaling. **[HEURISTIC]**

F. Daily Picks / finite deck

User goal: calm, intentional browsing; avoid infinite swiping.

Capability stack:

- Deterministic deck generation.
- Optional LLM intro copy (repo already emphasizes finite/intentional/premium, bilingual). **[VERIFIED]** ⁷

Personality rule:

- No hidden personality inference.
- If personality is used, it must be an explicit toggle and included in "why these" provenance. **[INFERRED]**

Trust risks: covert ranking.

Privacy risks: using sensitive traits in recommender without disclosure.

Free vs premium:

- Free: standard deck
- Premium: “quality tooling” (filters, reflection prompts) rather than more swipes. **[HEURISTIC]**

G. “Why this match” explanations

User goal: understand why they’re seeing someone without feeling manipulated.

Capability stack:

- Structured outputs + strict allowlist of signals (repo already enforces whitelisted signals + no private preferences leakage). **[VERIFIED]** 5 15
- Add provenance ledger in output: “Used: X, Y, Z. Not used: personality unless disclosed.” **[INFERRED]**

Fast vs deep: Fast (low thinking), because it is explanatory, not therapeutic.

Trust risks: “AI decided” vibes.

Privacy risks: revealing private taste profile (repo excludes private preferences). **[VERIFIED]** 41

H. Messaging assistance

User goal: write respectful, authentic messages; avoid escalation and cringe.

Capability stack:

- Rewrite-first mode (must start from user draft).
- Structured output message_coaching_result; no auto-send; no impersonation. **[VERIFIED]** repo’s policies emphasize drafts/author control and no autosend. 5

Fast vs deep:

- Default: low latency (Flash / low thinking). **[VERIFIED]** Flash models positioned for low latency. 42
- “Think deeper” button: high thinking for conflict-sensitive messages (premium). **[VERIFIED]** thinking controls exist. 22

Trust risks: “performed romance” (overly polished intimacy).

Privacy risks: storing chat text; must be ephemeral. **[HEURISTIC]**

I. Safety center

User goal: learn how to stay safe; understand what the app will do/not do.

Capability stack:

- Search grounding with citations for safety Q&A (fresh info); Gemini grounding returns structured citation metadata. **[VERIFIED]** 20
- Safety settings configured appropriately. **[VERIFIED]** 43

Visibility: Visible; citations required.

Trust risks: hallucinated safety advice; grounding mitigates. **[VERIFIED]** grounding provides sources/citations metadata. 20

Privacy risks: none if no personal data is inserted into the question.

Free vs premium:

- Always free.

J. Date planning

User goal: plan a safe first date easily; find midpoint venues.

Capability stack:

- Google Maps grounding (not Search) for venue lists + citations + optional widget context tokens. **[VERIFIED]** Maps grounding tool, metadata, and attribution requirements exist. 16
- Structured outputs if model supports “structured outputs with tools;” otherwise two-pass solution. **[VERIFIED]** structured outputs exist; and tools+structured outputs may be model-limited. 44

Visibility: Visible; citations UI required.

Fast vs deep: Medium depth; planning benefits from some reasoning, but latency should be managed.

Trust risks: suggesting unsafe/private venues; must enforce public venues.

Privacy risks: exact location; must use coarse location only (repo already states “coarse locations”).

[VERIFIED] 41

Free vs premium:

- Free: 2–3 venues
- Premium: “concierge itinerary” + backup venues + messaging templates for coordination. **[HEURISTIC]**

K. Premium concierge

User goal: calm, high-touch support (planning + reflection + communication).

Capability stack:

- High thinking + structured outputs

- Optional Live API voice reflection journaling later (opt-in). **[VERIFIED]** Live tool use support and limitations are documented. ³⁴

Trust risks: pseudo-therapy.

Privacy risks: audio storage, retention.

Recommended: “Concierge is coaching + planning + reflection; never diagnosis.” **[INFERRED]**

L. Internal moderation / support / eval ops

User goal: detect scams/harassment; triage reports; improve system.

Capability stack:

- Safety scan classifier outputs with schema (repo already has safety_scan schema). **[VERIFIED]** ³⁸
- Logs/datasets only with strict minimization; note retention statements. **[VERIFIED]** logs retention/ expiry and dataset creation are documented. ³¹
- Interactions API storage control (store=false if ever used). **[VERIFIED]** Interactions retention and store flag behavior exist. ³³

Governance, Trust, and Monetization

Prototype vs Production Boundary

Prototype-safe inside AI Studio Build mode:

- UI iteration via Build mode + annotation mode + AI chips. **[VERIFIED]** build/annotation/gallery/remix are documented. ²⁴
- Non-sensitive demo data only (synthetic users). **[HEURISTIC]**
- Quick deploy to Cloud Run for demos (still treat as staging). **[VERIFIED]** build mode supports Cloud Run deploy. ²⁴

Must move to GitHub + server runtime + Firebase AI Logic (before real users):

- Personality scoring service (deterministic, versioned). **[HEURISTIC]**
- Provenance ledger enforcement + whitelisted-signal gate for “why this match.” **[INFERRED]** aligns with Keshet’s existing policy pattern. ⁵
- App Check enforcement and per-user quotas if calling models from client via Firebase AI Logic. **[VERIFIED]** AI Logic supports App Check and per-user quotas. ⁴⁵

Fine in demos but too risky in production (unless heavily constrained):

- Interactions API with default storage (sensitive retention). **[VERIFIED]** store=true default retention described. ³³
- Logging entire prompts/responses for real users (retention + optional sharing). **[VERIFIED]** ³¹
- Image generation “icebreakers” (tone drift, misuse) unless tightly gated as labs (repo already demotes to labs). **[VERIFIED]** ⁴¹ ⁴⁶

Personality Data Governance Model

Minimum viable governance model:

- Data classes:
 - 1) **Raw responses** (most sensitive)
 - 2) **Computed scores** (aspects/domains)
 - 3) **User-facing reflections** (text cards)
 - 4) **Sharing permissions** (what's public to matches)

Storage rules:

- Store raw responses only if needed for export/retake; if stored, encrypt and separate from identity record. **[HEURISTIC]**
- Store computed aspect scores + version + timestamp as the default “personality profile.” **[INFERRED]** supports user value while minimizing retained sensitive content.
- “Reflection text” should be regenerable and not required to persist; if stored, store last N cards only or store templates + variables. **[HEURISTIC]**

User controls:

- View, edit profile labels, reset, export, delete (non-negotiable). **[HEURISTIC]**
- “Sharing controls” must be granular:
 - share nothing
 - share 1–2 aspect cards
 - share full aspect profile
 - share only conversation prompts (no scores)

What must never leak into ranking logic:

- Any “private reflection” text or safety flags. **[HEURISTIC]**
- Any personality signal not explicitly opted into for matching. **[INFERRED]**

Trust, Disclosure, and Anti-Chatfishing Guardrails

Guardrail set (hard rules):

- Rewrite-first as default: messaging assist requires user draft; output is options + rationale; never auto-send. **[VERIFIED]** repo's policies specify no autosend + user remains sender. 5
- No identity invention: every personal claim must be traceable to user input or an explicit allowed system signal. **[INFERRED]** implement via provenance ledger.
- Visible disclosure:
 - “AI suggestions” badge in composer (not in chat transcript by default)
 - optional “assisted” label per-message if user chooses (trust-forward option) **[HEURISTIC]**
- Anti-escalation nudges:
 - If user sends a high-intensity message early, offer “soften tone” suggestions.
 - Keep it optional; never block unless it's safety policy. **[HEURISTIC]**

- Safety defer: model recommends actions; deterministic policy + human review decide punitive action. **[VERIFIED]** safety settings + classifier schema patterns exist; and repo's safety_scan is for warnings/triage. 43 38

Feature Prioritization Matrix

Each feature is classified into exactly one category.

Feature	Category	Rationale	Evidence label
Deterministic BFAS scoring + user-owned personality profile	MUST BUILD NOW	Core differentiator; must be non-LLM for trust. BFAS structure established. 1	[INFERRED]
Personality reflection report (structured cards + uncertainty)	MUST BUILD NOW	High user value, low creep when framed as self-insight.	[INFERRED]
"Why this match" with provenance ledger	MUST BUILD NOW	Trust-forward core loop; repo already enforces whitelists. 5	[INFERRED]
Rewrite-first messaging coach	MUST BUILD NOW	Direct conversion lift; low-latency feasible; matches repo stance. 5 42	[INFERRED]
Maps-grounded date planner + citations	STRONG POST-MVP BET	Strong Google-native wedge; requires citations UI and safety constraints. 16	[INFERRED]
Compatibility reflection report (mutual opt-in)	STRONG POST-MVP BET	Valuable but easy to overclaim; must be careful with uncertainty and consent. 47	[INFERRED]
Daily Picks intro + "finite deck" tone	STRONG POST-MVP BET	Already in repo; helps positioning; low risk. 7	[VERIFIED]
Private taste profile assistant	LATER EXPERIMENT	Medium creep risk; requires strong user control and disclosure. Repo marks consent-required. 41	[INFERRED]
Image analysis for profile coaching	INTERNAL OPS ONLY	Too easy to drift into forbidden inference; reserve for moderation checks. 21	[INFERRED]
Live API voice reflection journal	LATER EXPERIMENT	High delight potential, high privacy risk; later-phase premium only. 34	[INFERRED]
Image generation icebreakers	LATER EXPERIMENT	Fun but distracts; keep "labs off by default" (repo already does). 41 46	[VERIFIED]

Feature	Category	Rationale	Evidence label
Moderation summarizer	INTERNAL OPS ONLY	Operational leverage; never member-facing. Repo already internal. ⁴¹	[VERIFIED]
Hidden compatibility scoring / destiny claims	DO NOT BUILD	Violates non-negotiables and outruns evidence.	[HEURISTIC]
Protected-trait inference from photos	DO NOT BUILD	Disallowed by your rules; also high harm risk.	[HEURISTIC]

Free vs Premium Boundary

Principle: monetize **calm depth and convenience**, not access to dignity, safety, or basic self-knowledge.

Free (baseline dignity):

- Full personality assessment + basic reflection report (short).
- Basic bio coach drafts (Hebrew-first).
- Basic “why this match” explanation with provenance.
- Safety center Q&A with citations.

Premium (depth + concierge):

- Expanded reflection modules (“relationship micro-habits,” “conflict style scripts”).
- “Think deeper” messaging for sensitive situations (high thinking).
- Date concierge: midpoint planning, backup venues, scheduling templates (Maps grounded).
- Optional voice reflection journal (later).
- Priority human support escalation.

Evidence: Pricing strategy is product judgment, not validated by the cited sources. **[HEURISTIC]**

Schemas, Prototypes, and Production Architecture

Exact JSON Schemas

Below are concrete JSON Schemas (Gemini structured output supports a subset of JSON Schema; keep schemas simple: types, properties, required, enums, arrays, nullables). **[VERIFIED]** ¹⁷

Global conventions (apply to all objects):

- `schema_version`: required string
- `generated_at`: required ISO-8601 string
- `language`: required BCP-47 string (e.g., `"en-US"`, `"he"`)
- `confidence`: number 0–1
- `evidence_label`: enum `["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"]`
- Unknowns: use `null` plus optional `unknown_reason`

Note: These schemas are designed both for (a) UI objects and (b) logging-safe structured outputs. Sensitive raw inputs should not be embedded unless required.

```
{
  "$id": "https://kesher.ai/schemas/uncertainty_marker.json",
  "type": "object",
  "properties": {
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": {
      "type": "string",
      "enum": ["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"]
    },
    "unknown_reason": { "type": ["string", "null"] },
    "notes": { "type": ["string", "null"] }
  },
  "required": ["confidence", "evidence_label"]
}
```

```
{
  "$id": "https://kesher.ai/schemas/provenance_ledger.json",
  "type": "object",
  "properties": {
    "ledger_version": { "type": "string" },
    "entries": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "field_path": { "type": "string", "description": "JSON pointer-like path, e.g. /reasons/0" },
          "source_type": {
            "type": "string",
            "enum": ["user_input", "candidate_input", "user_setting", "system_policy", "grounding_maps", "grounding_search", "deterministic_score"]
          },
          "source_id": { "type": ["string", "null"], "description": "Opaque ID (not raw content)" },
          "source_excerpt_hash": { "type": ["string", "null"], "description": "Hash of snippet, not snippet itself" },
          "allowed_for_user_display": { "type": "boolean" }
        },
        "required": ["field_path", "source_type", "allowed_for_user_display"]
      }
    }
  },
}
```

```
    "required": ["ledger_version", "entries"]
}
```

```
{
  "$id": "https://kesher.ai/schemas/risk_flags.json",
  "type": "object",
  "properties": {
    "creepiness_risk": { "type": "string", "enum": ["low", "medium", "high"] },
    "privacy_risk": { "type": "string", "enum": ["low", "medium", "high"] },
    "manipulation_risk": { "type": "string", "enum": ["low", "medium",
"high"] },
    "stereotype_risk": { "type": "string", "enum": ["low", "medium", "high"] },
    "needs_human_review": { "type": "boolean" },
    "notes": { "type": ["string", "null"] }
  },
  "required": ["creepiness_risk", "privacy_risk", "manipulation_risk",
"stereotype_risk", "needs_human_review"]
}
```

```
{
  "$id": "https://kesher.ai/schemas/personality_profile_summary.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "domains": {
      "type": "object",
      "properties": {
        "extraversion": { "type": ["number", "null"] },
        "neuroticism": { "type": ["number", "null"] },
        "agreeableness": { "type": ["number", "null"] },
        "conscientiousness": { "type": ["number", "null"] },
        "openness_to_experience": { "type": ["number", "null"] }
      },
      "required":
["extraversion", "neuroticism", "agreeableness", "conscientiousness", "openness_to_experience"]
    },

    "scale": {
      "type": "object",
```



```

    "properties": {
      "score_type": { "type": "string", "enum": ["z_score", "percentile",
"raw_0_1"] },
      "norm_reference": { "type": ["string", "null"], "description": "e.g.,
'kesher_beta_2026Q2' or null if unknown" },
      "unknown_reason": { "type": ["string", "null"] }
    },
    "required": ["score_type"]
  },

  "high_level_takeaways": {
    "type": "array",
    "minItems": 2,
    "maxItems": 5,
    "items": { "type": "string" }
  },

  "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
  "provenance": { "$ref": "https://kesher.ai/schemas/
provenance_ledger.json" },
  "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
},
"required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "domains", "scale", "high
}

```

```

{
  "$id": "https://kesher.ai/schemas/aspect_profile_summary.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "aspects": {
      "type": "object",
      "properties": {
        "enthusiasm": { "type": ["number", "null"] },
        "assertiveness": { "type": ["number", "null"] },

        "withdrawal": { "type": ["number", "null"] },
        "volatility": { "type": ["number", "null"] },

```

```

    "compassion": { "type": ["number", "null"] },
    "politeness": { "type": ["number", "null"] },

    "industriousness": { "type": ["number", "null"] },
    "orderliness": { "type": ["number", "null"] },

    "openness": { "type": ["number", "null"] },
    "intellect": { "type": ["number", "null"] }
  },
  "required":
["enthusiasm", "assertiveness", "withdrawal", "volatility", "compassion", "politeness", "industriousness",
  ],

  "date_relevant_cards": {
    "type": "array",
    "minItems": 3,
    "maxItems": 8,
    "items": {
      "type": "object",
      "properties": {
        "title": { "type": "string" },
        "what_it_can_look_like_in_dating": { "type": "string" },
        "strength_if_managed_well": { "type": "string" },
        "watch_out": { "type": "string" },
        "try_this_micro_habit": { "type": "string" },
        "unknown_reason": { "type": ["string", "null"] }
      },
      "required":
["title", "what_it_can_look_like_in_dating", "strength_if_managed_well", "watch_out", "try_this_micro_habit",
    ]
    },
  },

  "uncertainty": { "$ref": "https://kesher.ai/schemas/uncertainty_marker.json" },
  "provenance": { "$ref": "https://kesher.ai/schemas/provenance_ledger.json" },
  "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
},
"required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "aspects", "date_relevant_cards"]
}

```

```

{
  "$id": "https://kesher.ai/schemas/compatibility_reflection_card.json",
  "type": "object",

```

```

"properties": {
  "schema_version": { "type": "string" },
  "generated_at": { "type": "string" },
  "language": { "type": "string" },
  "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
  "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

  "card_type": { "type": "string", "enum": ["possible_strength",
"possible_friction", "question_to_explore", "micro_habit"] },
  "title": { "type": "string" },
  "what_we_know": { "type": "string", "description":
"Only from shared inputs; no guessing" },
  "why_it_might_matter": { "type": "string" },
  "suggested_question": { "type": "string" },
  "gentle_boundary": { "type": "string" },
  "unknown_reason": { "type": ["string", "null"] },

  "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
  "provenance": { "$ref": "https://kesher.ai/schemas/
provenance_ledger.json" },
  "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
},
"required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "card_type", "title", "wh
}

```

```

{
  "$id": "https://kesher.ai/schemas/why_this_match_card.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "reasons": { "type": "array", "minItems": 2, "maxItems": 3, "items": {
"type": "string" } },
    "first_question": { "type": "string" },
    "possible_mismatch_to_clarify": { "type": ["string", "null"] },

    "signals_used": { "type": "array", "items": { "type": "string" } },
    "signals_not_used": { "type": "array", "items": { "type": "string" } },

```

```

    "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
    "provenance": { "$ref": "https://kesher.ai/schemas/
provenance_ledger.json" },
    "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
  },
  "required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "reasons", "first_questi
}

```

```

{
  "$id": "https://kesher.ai/schemas/daily_picks_intro.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "headline_en": { "type": "string" },
    "headline_he": { "type": "string" },
    "body_en": { "type": "string" },
    "body_he": { "type": "string" },

    "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
    "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
  },
  "required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "headline_en", "headline
}

```

```

{
  "$id": "https://kesher.ai/schemas/message_coaching_result.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

```

```

    "mode": { "type": "string", "enum": ["rewrite_only", "light_options",
"explain_only"] },
    "user_intent_summary": { "type": "string" },
    "drafts": {
      "type": "array",
      "minItems": 2,
      "maxItems": 4,
      "items": {
        "type": "object",
        "properties": {
          "text": { "type": "string" },
          "tone_label": { "type": "string" },
          "what_changed": { "type": "string" }
        },
        "required": ["text","tone_label","what_changed"]
      }
    },
    "do_not_send_warning": { "type": ["string","null"], "description":
"Optional nudge if too intense / unsafe" },

    "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
    "provenance": { "$ref": "https://kesher.ai/schemas/
provenance_ledger.json" },
    "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
  },
  "required":
["schema_version","generated_at","language","confidence","evidence_label","mode","user_intent_sum
}

```

```

{
  "$id": "https://kesher.ai/schemas/values_phrasing_result.json",
  "type": "object",
  "properties": {
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    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED","INFERRED","HEURISTIC","UNKNOWN"] },

    "phrasing_options_he": { "type": "array", "minItems": 3, "maxItems": 6,
"items": { "type": "string" } },
    "phrasing_options_en": { "type": "array", "minItems": 3, "maxItems": 6,
"items": { "type": "string" } },

```

```

    "what_each_option_signals": { "type": "array", "items": { "type":
"string" } },
    "avoid_phrases": { "type": "array", "items": { "type": "string" } },

    "uncertainty": { "$ref": "https://kesher.ai/schemas/
uncertainty_marker.json" },
    "provenance": { "$ref": "https://kesher.ai/schemas/
provenance_ledger.json" },
    "risk_flags": { "$ref": "https://kesher.ai/schemas/risk_flags.json" }
  },
  "required":
["schema_version", "generated_at", "language", "confidence", "evidence_label", "phrasing_options_he", "
}

```

```

{
  "$id": "https://kesher.ai/schemas/date_plan_result.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "coarse_area_label": { "type": "string" },
    "time_window": { "type": ["string", "null"] },
    "venues": {
      "type": "array",
      "minItems": 3,
      "maxItems": 6,
      "items": {
        "type": "object",
        "properties": {
          "name": { "type": "string" },
          "place_id": { "type": ["string", "null"] },
          "why_good": { "type": "string" },
          "safety_note": { "type": "string" },
          "meetup_script": { "type": "string" },
          "sources": {
            "type": "array",
            "items": {
              "type": "object",
              "properties": {
                "title": { "type": "string" },
                "uri": { "type": "string" }

```

```

        },
        "required": ["title","uri"]
    }
}
},
"required": ["name","why_good","safety_note","meetup_script","sources"]
}
},
"uncertainty": { "$ref": "https://keshher.ai/schemas/
uncertainty_marker.json" },
"risk_flags": { "$ref": "https://keshher.ai/schemas/risk_flags.json" }
},
"required":
["schema_version","generated_at","language","confidence","evidence_label","coarse_area_label","ve
}

```

```

{
  "$id": "https://keshher.ai/schemas/safety_classification_result.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
["VERIFIED","INFERRED","HEURISTIC","UNKNOWN"] },

    "risk_level": { "type": "string", "enum":
["low","medium","high","uncertain"] },
    "categories": {
      "type": "array",
      "items": { "type": "string", "enum":
["harassment","hate","sexual","danger","scam","none"] }
    },
    "recommended_action": {
      "type": "string",
      "enum":
["allow","warn","user_nudge","block_and_report","needs_human_review"]
    },
    "short_rationale": { "type": "string" },

    "uncertainty": { "$ref": "https://keshher.ai/schemas/
uncertainty_marker.json" },
    "risk_flags": { "$ref": "https://keshher.ai/schemas/risk_flags.json" }
  },
}

```

```

    "required":
    ["schema_version", "generated_at", "language", "confidence", "evidence_label", "risk_level", "categories"]
  }

```

```

{
  "$id": "https://keshher.ai/schemas/personality_data_disclosure.json",
  "type": "object",
  "properties": {
    "schema_version": { "type": "string" },
    "generated_at": { "type": "string" },
    "language": { "type": "string" },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "evidence_label": { "type": "string", "enum":
    ["VERIFIED", "INFERRED", "HEURISTIC", "UNKNOWN"] },

    "plain_language_summary": { "type": "string" },
    "what_we_store": { "type": "array", "items": { "type": "string" } },
    "what_we_do_not_store": { "type": "array", "items": { "type": "string" } },
    "how_matching_uses_personality": { "type": "string" },
    "user_controls": { "type": "array", "items": { "type": "string" } },
    "unknowns": { "type": "array", "items": { "type": "string" } },

    "uncertainty": { "$ref": "https://keshher.ai/schemas/
uncertainty_marker.json" },
    "risk_flags": { "$ref": "https://keshher.ai/schemas/risk_flags.json" }
  },
  "required":
  ["schema_version", "generated_at", "language", "confidence", "evidence_label", "plain_language_summary"]
}

```

Best AI Studio Build-Mode Prototype

Prototype goal: prove the wedge end-to-end with minimal moving parts:

- 1) BFAS intake (short demo subset) → deterministic scoring → personality reflection cards
- 2) “Why this match” card with provenance ledger
- 3) Maps-grounded date plan with citations UI stub
- 4) Rewrite-first messaging coach

How to do it quickly in AI Studio:

- Start from Build mode and remix/export (AI Studio supports remixing from App Gallery and exporting to GitHub; it also supports deploying to Cloud Run). **[VERIFIED]** ²⁴
- Use AI Chips to add “Google Maps data”/image generation only for prototypes (but keep image generation off by default). **[VERIFIED]** ²⁴
- Keep all outputs strictly structured (response JSON schema). **[VERIFIED]** ¹⁷

Repo alignment:

- The provided repo already follows a server-side proxy pattern (“all Gemini SDK usage is server-side only”), enforces a feature registry allowlist, and uses response schemas for several features.

[VERIFIED] 6 7 41

Best Production Architecture

Target: production-safe, trust-forward, Hebrew-first product for Israel 48 audiences (repo prompts explicitly reference Israel and Hebrew-first outputs). [VERIFIED] 15

Recommended architecture (hybrid, contract-first):

- Client (iOS/Android/Web):
 - BFAS UI (deterministic)
 - Messaging composer with rewrite-first
 - Date planning UI with citation widgets
- Backend (Cloud Run or equivalent):
 - **AI Gateway**: single endpoint; injects system policies; strips disallowed inputs; enforces schema; manages tool-use.
 - **Policy Engine** (deterministic): safety actions, blocklists, rate limits, audit logs.
 - **Personality Service** (deterministic): BFAS scoring, versioning, export/delete.
- Firebase layer:
 - Firebase Auth (not sourced here; implementation choice) [HEURISTIC]
 - Firebase AI Logic only for lower-risk client calls if used; *must* be protected with App Check. [VERIFIED] AI Logic + App Check integration is explicit. 29
 - Per-user AI quotas via Firebase AI Logic API quota controls. [VERIFIED] 27
 - Remote Config for feature gating and model routing *without storing confidential data* (Firebase explicitly warns against storing confidential data in Remote Config). [VERIFIED] 28

Logging posture:

- Disable broad prompt logging for real member content; if logs are needed, redact and sample. [INFERRED] Gemini logs are developer-owned, retained/expired, and can be shared via datasets if opted-in. 31

Model / Tool Routing Recommendations

Routing principles (Google-native):

- Use Structured Outputs whenever you need UI stability. [VERIFIED] 17
- Use Thinking controls:

- Low thinking for onboarding, “why this match,” quick rewrites. **[VERIFIED]** 22
- High thinking for personality reflection and compatibility. **[VERIFIED]** 22
- Use Maps grounding only when geographical intent is explicit; show sources immediately after content and within one interaction (Maps requirements). **[VERIFIED]** 16
- Use Search grounding for safety center Q&A and other freshness-required explanations; build citations using groundingSupports/groundingChunks metadata. **[VERIFIED]** 20

Suggested per-surface routes (model names are placeholders; select from your allowed registry):

- Messaging rewrite: Flash / low thinking
- Personality reflection: Pro / high thinking + schema
- Date planning: Maps-capable model + googleMaps tool
- Safety scan: faster classifier model + strict schema + conservative thresholds (block low and above, depending on sensitivity). **[VERIFIED]** threshold controls exist. 43

Evaluation and Red-Team Plan

Evaluation must test “trust failures,” not just model quality.

Plan:

- Build datasets from synthetic and consented logs only (avoid real sensitive chats in datasets). **[INFERRED]** Logging/datasets tooling exists and has retention limits and opt-in sharing controls. 31
- Red-team suites:
- Personality: “does it diagnose?”, “does it moralize traits?”, “does it assert certainty?” **[HEURISTIC]**
- Compatibility: “does it output a score?”, “does it claim destiny?”, “does it reveal the other person’s private traits?” **[INFERRED]** mirror the repo’s “whitelisted signals only” design intent. 5
- Safety scan: stress test categories and thresholds. **[VERIFIED]** safety categories/thresholds are defined. 43
- Grounding: verify citations render and match content. **[VERIFIED]** grounding metadata supports citations. 49

Operational metrics:

- % of “unknown” outputs (should be non-zero) **[HEURISTIC]**
- Citation coverage rate for grounded outputs **[HEURISTIC]**
- Human override rate in moderation flows **[HEURISTIC]**

UI / UX Concepts

UX should feel like: *editing + coaching + reflection + concierge + guardrail*, not “AI therapist.”

Concepts:

- “Personality is a lens” card at top of reports (one sentence + uncertainty language). **[INFERRED]** aligns with non-determinism implied by relationship evidence. 2
- Aspect cards with: “What it can look like,” “Strength,” “Watch-out,” “Try this.” (schema above)
- “Why this match” shows:

- 2–3 reasons
- 1 question
- “Signals used” and “Signals not used” (explicit provenance)
- Hebrew-first typography:
- RTL layout
- Avoid mixed-direction punctuation bugs by isolating Hebrew blocks **[HEURISTIC]**

Risks and Anti-Patterns

High-risk anti-patterns (do not build):

- Compatibility destiny claims or a single scalar “match score” that implies deterministic prediction. **[HEURISTIC]**
- Hidden use of personality data in ranking without disclosure/control. **[HEURISTIC]**
- Any inference of protected traits (religiosity, ethnicity, politics) from photos or writing style. **[HEURISTIC]**
- “AI performed romance”: autosend, impersonation, intimate roleplay. **[VERIFIED]** repo already forbids autonomy and identity invention in system instructions; preserve this principle. ⁵
- Using Interactions API with store=true for sensitive dating chats by default (retention risk). **[VERIFIED]** Interactions stored by default with stated retention windows unless store=false. ³³
- Over-logging prompts and later contributing datasets with sensitive user content. **[VERIFIED]** logs/datasets + sharing controls exist; retention is described. ³¹

Final Recommendation

Build the wedge as a **trust-forward personality layer**:

- 1) Deterministic BFAS scoring + user-owned profile
- 2) Structured personality reflection cards with uncertainty markers
- 3) Provenance-driven “why this match”
- 4) Rewrite-first messaging coach (low latency)
- 5) Maps-grounded date concierge (post-MVP or MVP+ if team can ship attribution UI cleanly)

Treat everything else (taste profiling, voice journaling, image generation) as later-phase, explicitly opt-in, and never required for basic dignity.

This approach is scientifically defensible at the level the evidence supports (personality relates to relationship satisfaction, but not deterministically). **[VERIFIED]** ²

It is culturally respectful because it emphasizes user agency and avoids stereotyping and hidden inference (consistent with Kesher’s repo policies). **[VERIFIED]** ⁵

Prompt Pack and Operational Scoring

Copy-Paste Prompt Pack

Each pack includes: system instruction, base prompt, output schema ref, settings guidance, tool guidance, avoid list, prototype-safe notes, production-safe notes.

Hebrew-first bio coach

```
{
  "system_instruction": "You are Keshet's profile coach. Help the user express themselves authentically in Hebrew (RTL-aware), calm and respectful. Do not invent facts. Produce drafts only. No attractiveness scoring. No stereotypes about Judaism/observance/ethnicity/politics. The user remains the author; no auto-send.",
  "base_prompt": "User bio (raw): {{bio_raw}}\nTone: {{tone}}\nValues/observance: {{values}}\nDealbreakers (optional): {{dealbreakers}}\nLength: {{length}}\n\nReturn 3 Hebrew-first draft bios plus optional conversation hooks. Preserve user facts exactly.",
  "output_schema": { "$ref": "https://keshet.ai/schemas/message_coaching_result.json" },
  "settings_guidance": {
    "thinking": "low",
    "temperature": 0.3,
    "structured_output": true
  },
  "tool_guidance": { "tools": [] },
  "avoid_list": [
    "No invented life details",
    "No diagnosing personality or mental health",
    "No objectifying language",
    "No claims about user religiosity beyond provided text"
  ],
  "prototype_safe_notes": "Safe in AI Studio with synthetic bios. Use response_json_schema structured output.",
  "production_safe_notes": "Run server-side or via Firebase AI Logic with App Check; do not store raw bios in logs; keep drafts ephemeral until user saves."
}
```

Values / observance phrasing assistant

```
{
  "system_instruction": "You help the user phrase values/observance in a respectful, non-stereotyping way in both Hebrew and English. Do not infer identity. Offer multiple options with different levels of directness.",
  "base_prompt": "User intent: {{intent}}\nUser constraints: {{constraints}}\n\nAudience: serious dating\n\nGenerate phrasing options in Hebrew and English, plus what each option signals and phrases to avoid.",
  "output_schema": { "$ref": "https://keshet.ai/schemas/values_phrasing_result.json" },
  "settings_guidance": { "thinking": "medium", "temperature": 0.4,
    "structured_output": true },
  "tool_guidance": { "tools": [] },
}
```

```

    "avoid_list": ["No stereotyping groups", "No political assumptions", "No moralizing trait differences"],
    "prototype_safe_notes":
    "Works in AI Studio Build mode; validate bilingual RTL rendering.",
    "production_safe_notes": "Treat as rewrite/options; never auto-fill public profile without user confirmation."
  }

```

Personality reflection summary

```

{
  "system_instruction":
  "You are a personality reflection assistant for a dating app. Use the provided BFAS domain/aspect scores only. Do not diagnose. Do not predict relationship fate. Frame as tendencies with uncertainty markers. Output structured cards.",
  "base_prompt": "User BFAS scores:\nDomains: {{domains}}\nAspects: {{aspects}}\n\nScale metadata: {{scale}}\nLanguage: {{language}}\n\nGenerate date-relevant reflection cards: what it can look like, strengths, watch-outs, and a micro-habit. Use uncertainty language.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/aspect_profile_summary.json" },
  "settings_guidance": { "thinking": "high", "temperature": 0.4,
  "structured_output": true },
  "tool_guidance": { "tools": [] },
  "avoid_list": ["No diagnosis", "No certainty language", "No moralizing traits"],
  "prototype_safe_notes": "Use synthetic score vectors first; ensure the model does not hallucinate scores.",
  "production_safe_notes": "Scores must come from deterministic scorer; store provenance ledger; redact from logs."
}

```

Compatibility reflection card

```

{
  "system_instruction": "You generate a single compatibility reflection card. You may only use mutually shared inputs (explicitly provided). No compatibility score. No destiny language. Output one card with a suggested question and a gentle boundary.",
  "base_prompt": "Shared inputs (whitelisted only): {{whitelist_bundle}}\n\nCreate exactly one compatibility_reflection_card.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/compatibility_reflection_card.json" },
  "settings_guidance": { "thinking": "high", "temperature": 0.3,
  "structured_output": true },

```

```

    "tool_guidance": { "tools": [] },
    "avoid_list": ["No 'perfect match'", "No private inference", "No disallowed signals"],
    "prototype_safe_notes": "Use carefully constructed whitelist bundles; test failure cases with missing data (must return unknown_reason).",
    "production_safe_notes": "Require mutual opt-in flags; generate provenance ledger and store minimal audit metadata only."
}

```

Why-this-match explanation

```

{
  "system_instruction": "You are Kesher's match explainer. Use only whitelisted signals. Do not reveal private preferences or hidden ranking. Use respectful, probabilistic language. Output structured reasons + first question + signals used/not used.",
  "base_prompt": "User-visible signals: {{signals}}\nUser profile (visible fields only): {{user_profile_visible}}\nCandidate profile (visible fields only): {{candidate_profile_visible}}\n\nGenerate why_this_match_card.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/why_this_match_card.json" },
  "settings_guidance": { "thinking": "low", "temperature": 0.2,
"structured_output": true },
  "tool_guidance": { "tools": [] },
  "avoid_list": ["No hidden inference", "No certainty", "No mention of private taste profile"],
  "prototype_safe_notes": "Matches repo's existing design intent; easy to demo.",
  "production_safe_notes": "Enforce whitelist server-side; reject requests with non-allowed fields before model call."
}

```

Message rewrite assistant

```

{
  "system_instruction": "You are a rewrite-first communication coach. The user remains the author. You must keep the original intent and facts. Provide 2-4 rewritten options and what changed. No auto-send.",
  "base_prompt": "Original draft: {{user_draft}}\nDesired tone: {{tone}}\nContext (optional): {{context}}\n\nReturn message_coaching_result in rewrite_only mode.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/message_coaching_result.json" },
  "settings_guidance": { "thinking": "low", "temperature": 0.3,
"structured_output": true },

```

```

    "tool_guidance": { "tools": [] },
    "avoid_list": ["No flirting escalation beyond user intent", "No impersonation", "No invented details"],
    "prototype_safe_notes": "Use safe canned examples; test with Hebrew and English.",
    "production_safe_notes": "Do not log raw messages; store only aggregate usage metrics."
  }

```

Difficult-message “think deeper” assistant

```

{
  "system_instruction": "You help the user handle a difficult dating message thoughtfully. You must not provide therapy or diagnosis. Provide options, boundaries, and a short self-check. Keep it calm. User remains sender.",
  "base_prompt": "User draft: {{user_draft}}\nSituation: {{situation}}\nGoal: {{goal}}\n\nReturn message_coaching_result with light_options mode and a gentle boundary suggestion.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/message_coaching_result.json" },
  "settings_guidance": { "thinking": "high", "temperature": 0.35,
  "structured_output": true },
  "tool_guidance": { "tools": [] },
  "avoid_list": ["No therapy framing", "No diagnosing the other person", "No coercive advice"],
  "prototype_safe_notes": "Demo with synthetic scenarios; ensure uncertainty markers appear.",
  "production_safe_notes": "Add escalation path to human support for safety concerns."
}

```

Safe venue planner with Maps grounding

```

{
  "system_instruction": "You are a date-planning assistant. Prioritize safety (public venues), clarity, and user control. Use Google Maps grounding. Provide sources immediately after claims. Do not infer home addresses; use only coarse location.",
  "base_prompt": "Coarse area: {{coarse_area}}\nTime window: {{time_window}}\nPreferences: {{preferences}}\nBudget: {{budget}}\n\nReturn a date_plan_result with venues and sources.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/date_plan_result.json" },
  "settings_guidance": { "thinking": "medium", "temperature": 0.3,
  "structured_output": true },

```

```

    "tool_guidance": { "tools": ["googleMaps"], "enableWidget": true },
    "avoid_list": ["No private residences", "No unsafe venues", "No exact user
location storage"],
    "prototype_safe_notes":
"Requires Maps grounding tool; build citations UI from groundingChunks/
groundingSupports.",
    "production_safe_notes": "Follow Maps attribution requirements; store only
coarse area labels; never store lat/long unless necessary."
}

```

Safety-center cited answer flow

```

{
  "system_instruction": "You answer safety questions using Google Search
grounding. Provide concise, practical guidance. Include citations based on
grounding metadata. Do not include personal user data.",
  "base_prompt": "User safety question: {{question}}\nLocale: {{locale}}
\n\nAnswer with citations and suggest contacting support for urgent concerns.",
  "output_schema": { "type": "object", "properties": { "answer": { "type":
"string" } }, "required": ["answer"] },
  "settings_guidance": { "thinking": "low", "temperature": 0.2,
"structred_output": false },
  "tool_guidance": { "tools": ["googleSearch"] },
  "avoid_list": ["No hallucinated legal claims", "No medical advice", "No panic
language"],
  "prototype_safe_notes": "Grounding metadata can be rendered as inline
citations.",
  "production_safe_notes": "Do not combine with function calling if unsupported
in your tool stack; keep the question generic."
}

```

Harassment / scam classifier

```

{
  "system_instruction":
"You are a safety classifier. Output labels with minimal rationale. When
uncertain, mark uncertain and recommend human review. Do not moralize.",
  "base_prompt": "Message text: {{message_text}}\nContext (optional):
{{context}}\n\nReturn safety_classification_result.",
  "output_schema": { "$ref": "https://keshner.ai/schemas/
safety_classification_result.json" },
  "settings_guidance": { "thinking": "minimal", "temperature": 0.0,
"structred_output": true },
  "tool_guidance": { "tools": [] },
  "avoid_list": ["No personal attacks", "No speculative claims about identity"],
}

```



```

    "prototype_safe_notes": "Validate against known categories and edge cases;
    tune safety thresholds.",
    "production_safe_notes": "Use conservative safety settings and route high/
    uncertain to human review; deterministic enforcement decides outcomes."
}

```

AI disclosure / trust hub explainer

```

{
  "system_instruction": "You explain how Kesher uses AI in plain language. Be
  transparent, calm, and specific. Include user controls: view/edit/reset/export/
  delete. Never claim hidden profiling.",
  "base_prompt": "Product AI features list: {{features}}\nPersonality usage
  policy: {{personality_policy}}\nData retention summary: {{retention_summary}}
  \n\nReturn personality_data_disclosure.",
  "output_schema": { "$ref": "https://kesher.ai/schemas/
  personality_data_disclosure.json" },
  "settings_guidance": { "thinking": "medium", "temperature": 0.2,
  "structured_output": true },
  "tool_guidance": { "tools": [] },
  "avoid_list": ["No vague 'we may' language", "No implied surveillance", "No
  marketing hype"],
  "prototype_safe_notes": "Use consistent vocabulary across app surfaces.",
  "production_safe_notes": "Keep disclosures synchronized with actual system
  behavior; failing that is a trust-break."
}

```

Evaluation Rubric

Scores (1–10) for major recommended features.

Feature	Scientific validity	User value	Trust lift	Anti-creepiness	Privacy safety	Hebrew/RTL fit	Impl. feasibility	Prod. feasibility	Differentiation
Deterministic BFAS scoring + profile	9	8	9	9	7	6	7	7	8
Personality reflection report	8	9	8	8	7	7	7	7	8
Provenance “why this match”	6	8	10	9	8	8	8	8	9

Feature	Scientific validity	User value	Trust lift	Anti-creepiness	Privacy safety	Hebrew/RTL fit	Impl. feasibility	Prod. feasibility	Differentiation
Rewrite-first messaging coach	5	9	8	8	7	8	9	9	6
Maps-grounded date concierge	6	8	9	8	7	7	6	6	9
Compatibility reflection (mutual opt-in)	6	7	7	6	6	7	6	6	7
Safety center (Search grounded)	4	7	9	9	9	7	7	7	6

Overall stack scores:

- Trust-forward coherence: 9
- Product usefulness: 8
- Google-native leverage: 9
- Scientific integrity: 8
- Launch realism: 7
- Long-term defensibility: 8

Evidence label: these rubric scores are product judgments anchored to cited capabilities and literature, not directly validated numerically by the sources. **[HEURISTIC]**

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<https://pubmed.ncbi.nlm.nih.gov/17983306/>

2 39 <https://www.sciencedirect.com/science/article/abs/pii/S0092656609002001>

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3 40 47 <https://pubmed.ncbi.nlm.nih.gov/20718544/>

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4 17 44 <https://ai.google.dev/gemini-api/docs/structured-output>

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5 8 11 [policies.ts](https://github.com/akivagoldstein61-ui/Google-ai-studio/blob/main/src/ai/policies.ts)

<https://github.com/akivagoldstein61-ui/Google-ai-studio/blob/main/src/ai/policies.ts>

6 [server.ts](https://github.com/akivagoldstein61-ui/Google-ai-studio/blob/main/server.ts)

<https://github.com/akivagoldstein61-ui/Google-ai-studio/blob/main/server.ts>

7 aiRoutes.ts

<https://github.com/akivagoldstein61-ui/Google-ai-studio-/blob/main/server/aiRoutes.ts>

9 <https://www.understandmyself.com/>

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10 <https://scholarworks.gvsu.edu/orpc/vol4/iss4/1/>

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